

PrepBench Cheat Sheets

9 aptitude topics, one dense reference.

Percentage, Profit & Loss, Time & Work,
Speed-Distance-Time, Number System, Probability,
Squares & Cubes, Geometry, Data Interpretation.

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1. Percentage

Concept	Formula	Example
Percentage of a number	$x\% \text{ of } y = (x/100) \times y$	20% of 150 = 30
10% trick	divide by 10	10% of 450 = 45
5% trick	half of 10%	5% of 200 = 10
1% trick	divide by 100	1% of 500 = 5
Percentage change	$(\text{New}-\text{Old})/\text{Old} \times 100$	100 -> 120 = +20%

2. Profit & Loss

Concept	Formula	Example
Profit	$SP - CP$	CP 400, SP 500 -> 100
Loss	$CP - SP$	CP 500, SP 420 -> 80
Profit %	$(\text{Profit}/CP) \times 100$	100 on CP 400 = 25%
10% profit trick	$SP = (11/10) \times CP$	CP 500 -> SP 550
25% loss trick	$SP = (3/4) \times CP$	CP 400 -> SP 300

3. Time & Work

Concept	Formula	Example
Work	Time x Efficiency	5 days x 4 units/day = 20
One-day work	$1/x$ (finishes in x days)	10 days -> $1/10$ per day
Together (2 people)	$xy / (x+y)$	10d & 15d -> 6 days
Men-days-work	$M1D1 = M2D2$	$10 \times 12 = 15 \times 8$

4. Speed, Distance & Time

Concept	Formula	Example
Speed	Distance / Time	120km/2hr = 60 km/h
Distance	Speed x Time	60km/h x 4hr = 240 km
Average speed (equal dist.)	$2xy/(x+y)$	60 & 40 -> 48 km/h
Relative speed, same dir.	$ x - y $	80 & 60 -> 20 km/h
Relative speed, opposite dir.	$x + y$	60 & 40 -> 100 km/h

5. Number System

Rule	Test
Divisible by 2	last digit is 0,2,4,6,8
Divisible by 3	digit sum divisible by 3
Divisible by 4	last 2 digits divisible by 4
Divisible by 5	last digit is 0 or 5
Divisible by 9	digit sum divisible by 9
Divisible by 11	alternating digit sum is 0 or a multiple of 11
Remainder mod 9	= digit sum mod 9 (digital root shortcut)

6. Probability

Concept	Formula	Example
Probability	favorable / total	3 red of 10 -> 0.3
Complement	$1 - P(E)$	$P(E)=0.7 \rightarrow 0.3$
Independent events	$P(A) \times P(B)$	$1/2 \times 1/3 = 1/6$
Either A or B	$P(A)+P(B)-P(A \text{ and } B)$	$0.4+0.5-0.2 = 0.7$

7. Squares & Cubes

Concept	Formula	Example
Square	n^2	$12^2 = 144$
Cube	n^3	$5^3 = 125$
Numbers ending in 5	$(x5)^2 = x(x+1)$ then 25	$35^2 = 1225$
Difference of squares	$a^2 - b^2 = (a+b)(a-b)$	$15^2 - 5^2 = 20 \times 10 = 200$
Consecutive squares	$(n+1)^2 = n^2 + 2n + 1$	$11^2 = 10^2 + 21 = 121$

8. Geometry

Concept	Formula	Example
Circle ($\pi=22/7$)	use when radius is a multiple of 7	$r=7 \rightarrow$ exact fractions
Circle area	$\pi \times r^2$	$r=7 \rightarrow 154$
Circumference	$2 \times \pi \times r$	$r=7 \rightarrow 44$
Rectangle	Area = $l \times b$, Perimeter = $2(l+b)$	$l=8, b=5 \rightarrow 40, 26$
Pythagorean triplets	3-4-5, 5-12-13, 8-15-17	common right-triangle sides

9. Data Interpretation

Exam strategy (not formula-based)

Step	Why
Read the title first	know what the chart actually represents
Check units	%, Rs, lakhs, millions, crores all look similar at a glance
Read labels carefully	axes, legends, categories
Approximate before calculating	round values when exact numbers aren't required
Compare before calculating	estimate the answer, then refine only if needed
Percentage of total	$\% = (\text{category value} / \text{total of all categories}) \times 100$

prepbench (see the interactive tool for live practice questions)

Thank you for supporting PrepBench.